



Ingredient Safety & Product Composition Analysis

End-to-End Data Analytics Project
(Case Study 2)

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Data Analytics Bootcamp

Career target: **Safety Data Analyst –
Regulatory Affairs (Cosmetics & Fragrances)**





Context



-  The cosmetics industry is **highly regulated**
 -  Ingredients must comply with strict safety rules
 -  Ingredient compliance is critical for:
 - Consumer safety
 - Market access
 - Brand reputation
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Regulatory Background

What Is The European Cosmetic Regulation ?

- Regulation (EC) No 1223/2009** on cosmetic products
Ensures that Cosmetic products placed on the EU market are:
- Safe for human health
 - Comply with strict safety requirements



Regulatory Background

What Is the CosIng Database?

CosIng stands for: **Cos**metic **Ing**redients

Official European Commission Cosmetic
Ingredient Database

EC Regulation ▼

- All ▼
- All
 - Inventory (ingredient & fragrance)
 - Banned (Annex II)
 - Restricted (Annex III)
 - Colorants (Annex IV)
 - Preservatives (Annex V)
 - UV filters (Annex VI)



Regulatory Background

What Is a Restricted Ingredient?

Restricted ingredients are substances
allowed in cosmetics
But **only under specific conditions**

Maximum
concentration
limits

Restrictions to
certain product
types

Usage
warnings or
labeling
requirements



Regulatory Background

What Are CMR Substances?

CMR substances are chemicals classified as:

- C**arcinogenic (may cause cancer)
- M**utagenic (may cause genetic mutations)
- R**eprotoxic (may affect fertility or unborn children)

Generally **prohibited**

Exceptions exist **only**
under very strict
conditions

Objectives



Examine the prevalence of EU-restricted and potentially harmful substances (CMR) in cosmetic products available on the market



Key Questions:



- How are regulated ingredients distributed across cosmetic products, product categories and brands?
 - How do consumers respond to products containing these regulated ingredients?
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Data Sources

1. CosIng Annex III

Restricted cosmetic ingredients
CMR classification

rows: 375
columns: 16

2. Sephora Products

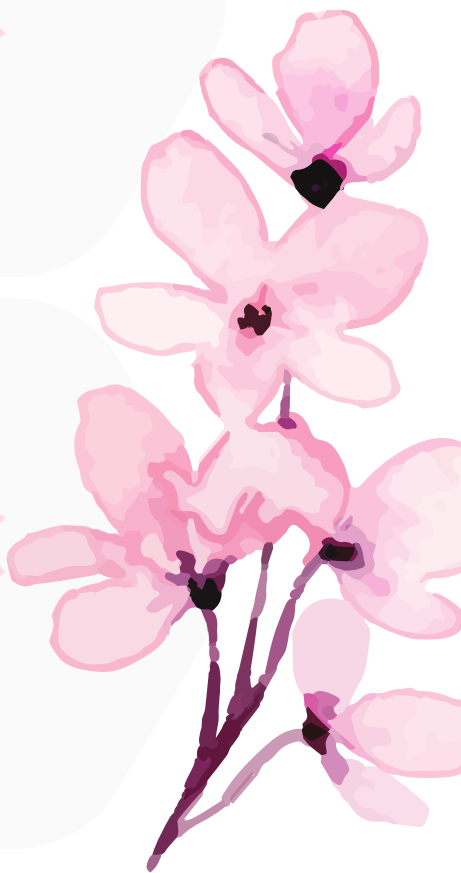
Major cosmetic product retailer
Full ingredient lists, ratings, categories...

rows: 8494
columns: 27

3. Skincare Products

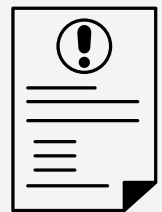
Additional Product and ingredient lists
coverage for skincare products

rows: 1472
columns: 11





Methodology



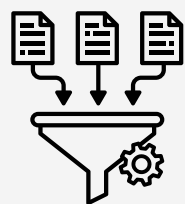
1

Data Cleaning & Harmonization



2

Cross-Reference



3

Quantitative Analysis





1.Data Cleaning & Harmonization

Normalize column names

Dropped rows where ingredients were missing

INGREDIENT MATCHING MUST BE CONSISTENT AND **EXACT**

Standardize ingredient names (UPPERCASE, strip spaces)

Unified separators as they vary: semicolons, line breaks..

Split ingredients strings into lists

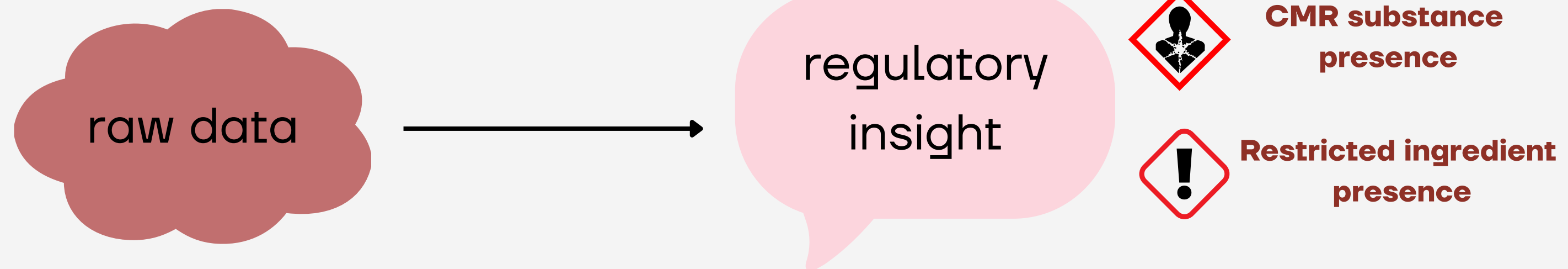
Exploded ingredient lists (ensured one ingredient per row)



2.Cross-Reference

Ingredient Matching Logic:

- ✓ Matched product ingredients against CosIng Annex III dataset
- ✓ Identified overlaps
- ✓ Linked each ingredient to regulatory status



3. Quantitative Analysis

Key Analytical Metrics:

Count of **restricted** ingredients

Count of **CMR** substances

Binary flags (has_cmr, has_restricted_ingredients) for each product

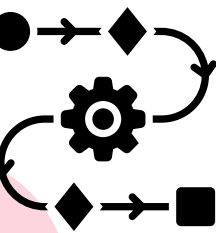
Percentage of restricted / CMR ingredients relative to the total ingredient count

Average consumer metrics such as price, rating, and engagement



Enabled product-level analysis

INGREDIENT
=
MEASURABLE
INDICATORS



Database Implementation



MySQL Database Structure

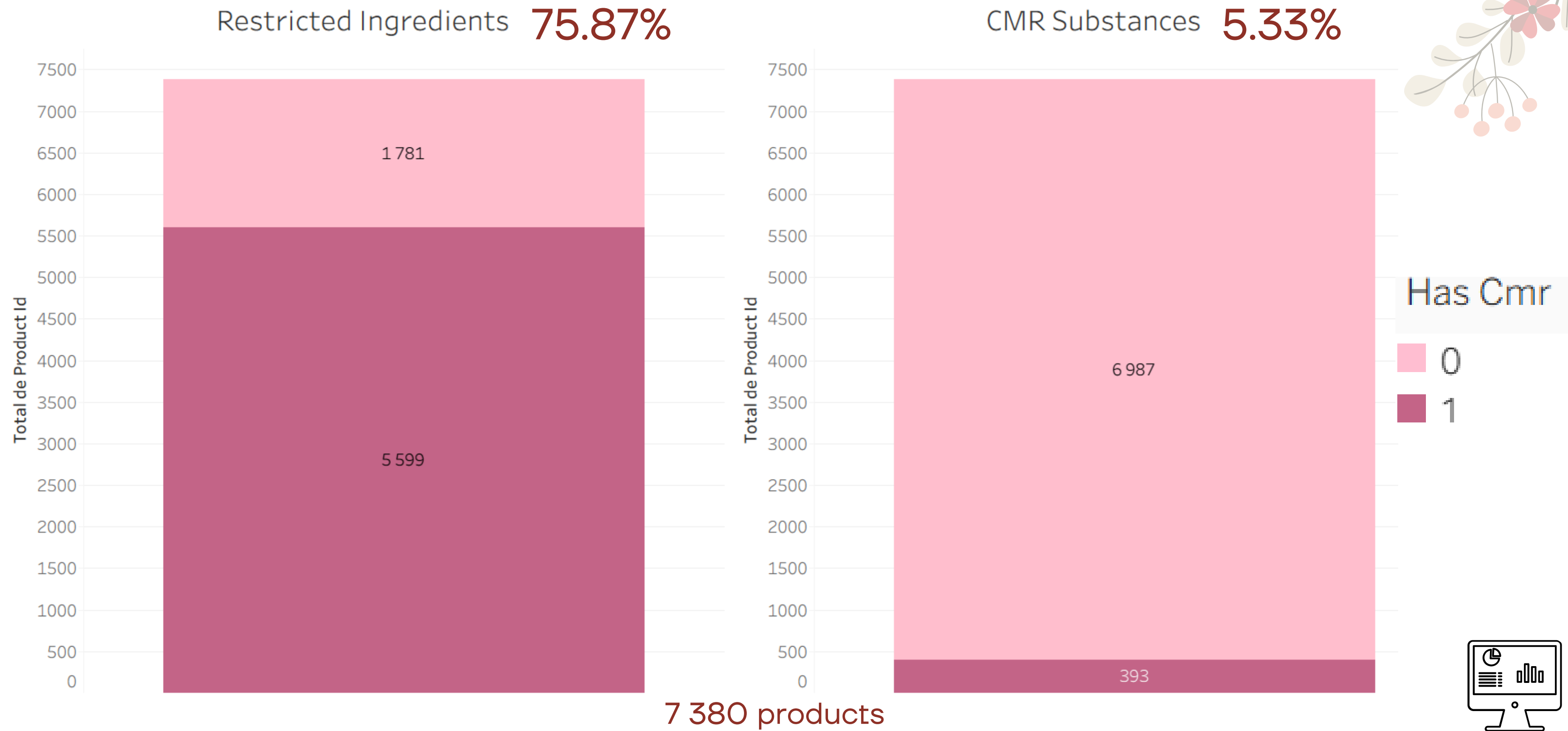
Tables created:

- sephora_products
- skincare_products

product_id	TEXT
product_name	TEXT
brand_name	TEXT
product_type	TEXT
price_usd	DOUBLE
rating	DOUBLE
reviews	DOUBLE
loves_count	BIGINT
restricted_ingredient_count	BIGINT
cmr_count	BIGINT
has_restricted_ingredient	BIGINT
has_cmr	BIGINT
restricted_pct	DOUBLE
cmr_pct	DOUBLE

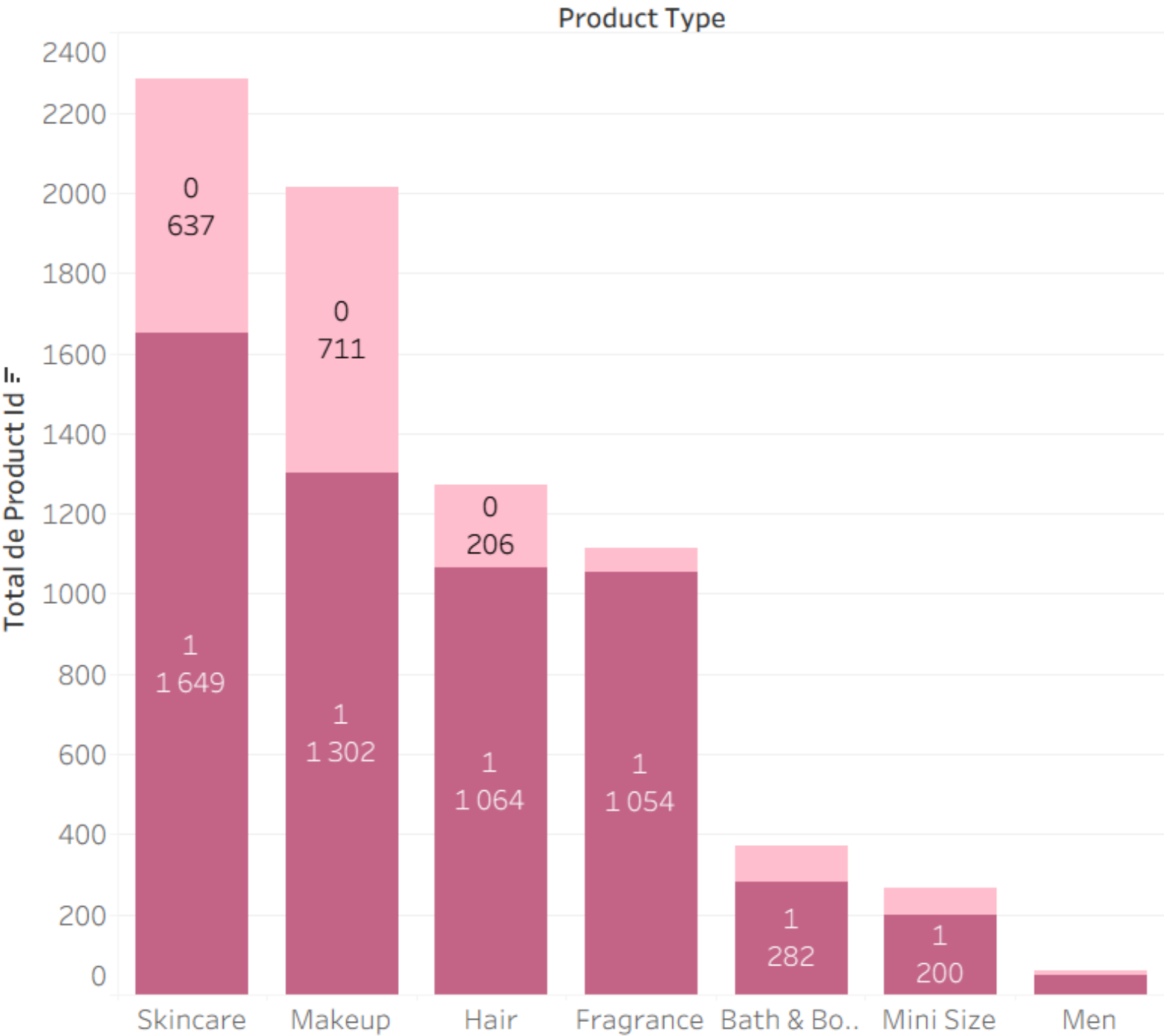
brand	TEXT
product_name	TEXT
product_type	TEXT
price	BIGINT
rating	DOUBLE
restricted_ingredient_count	BIGINT
cmr_count	BIGINT
has_restricted_ingredient	BIGINT
has_cmr	BIGINT
restricted_pct	DOUBLE
cmr_pct	DOUBLE

Regulated Ingredients Overview

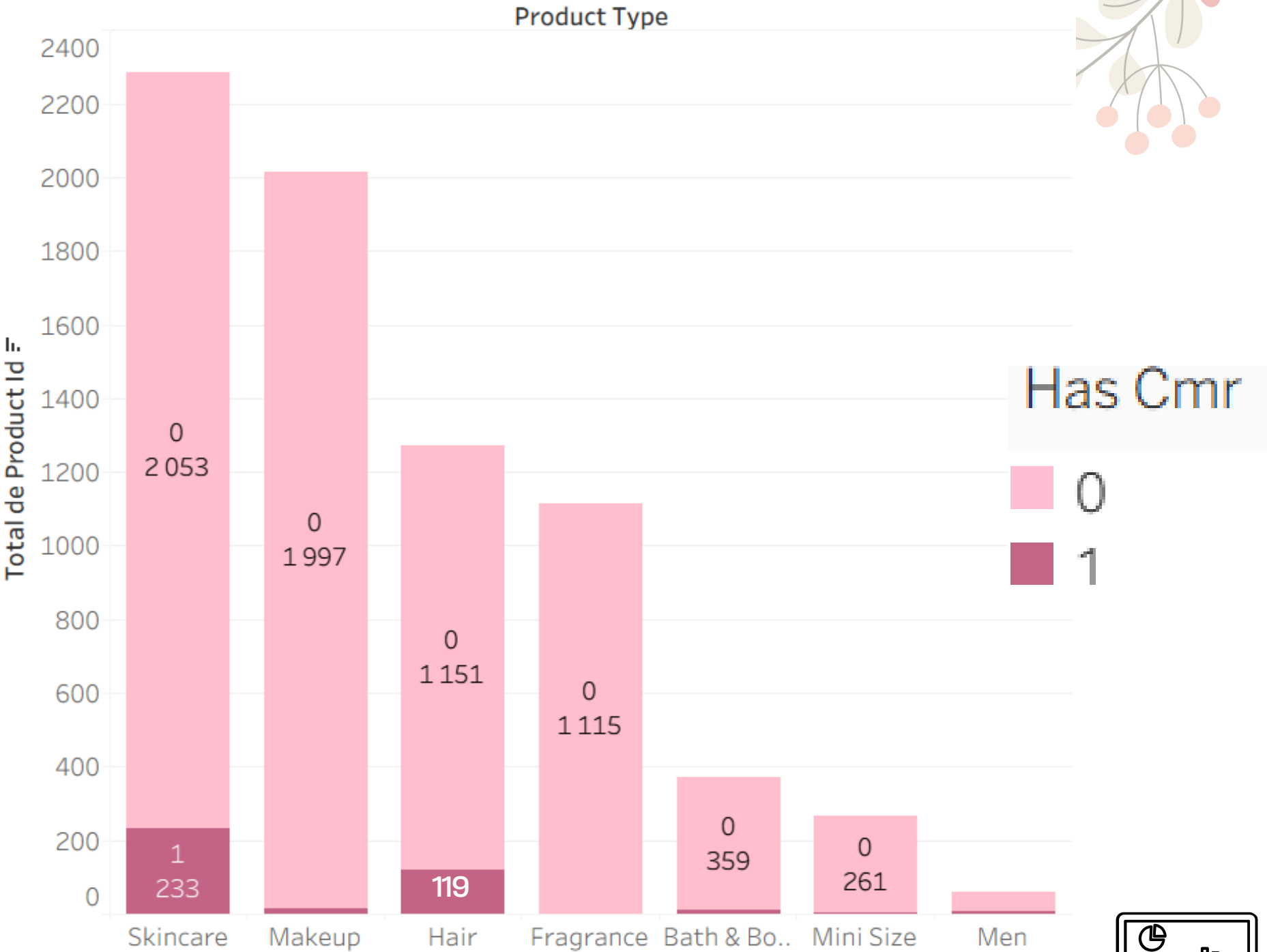


Product Types/Regulated Ingredients

Restricted Ingredients by Product Type

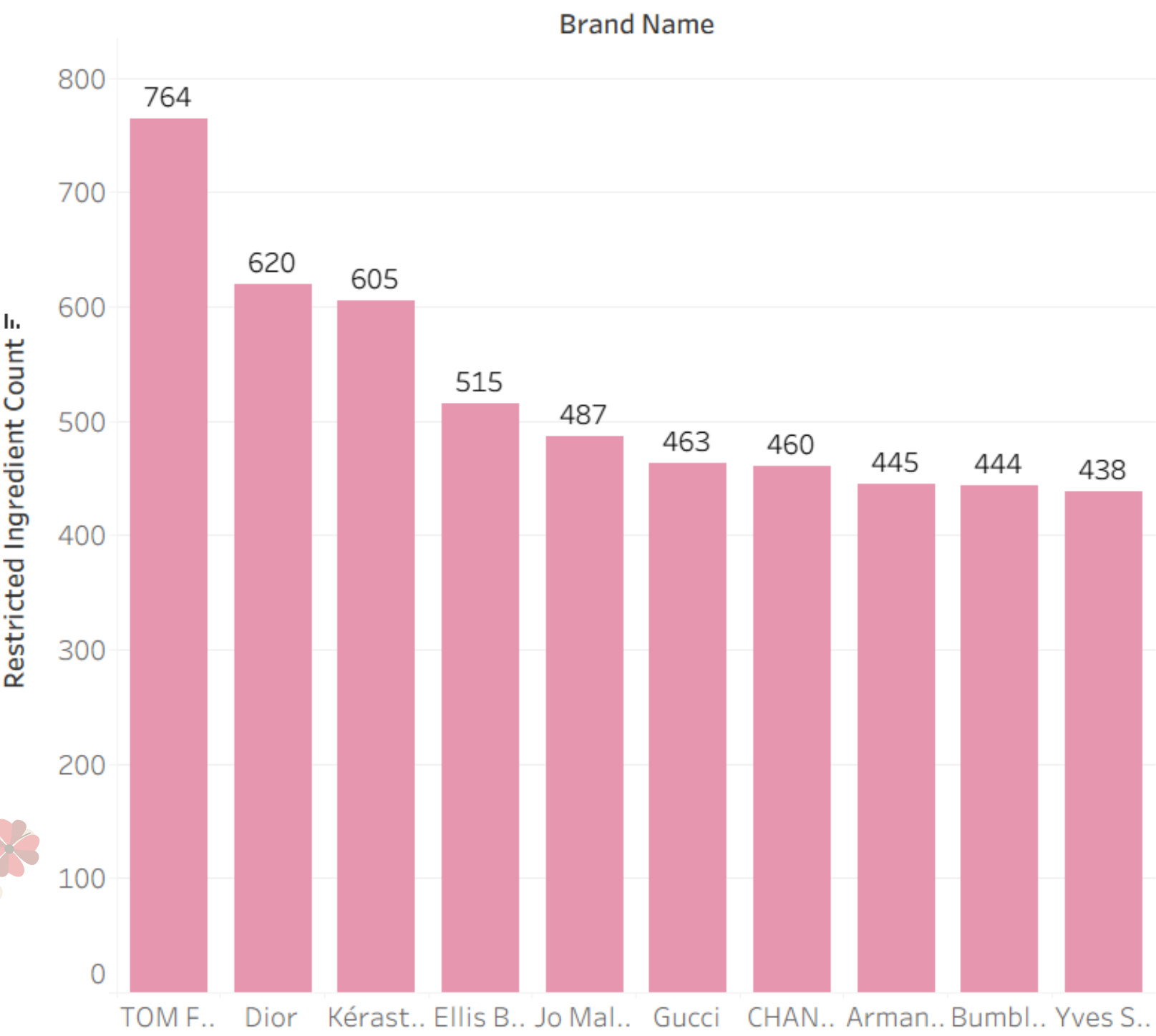


CMR Risk by Product Type

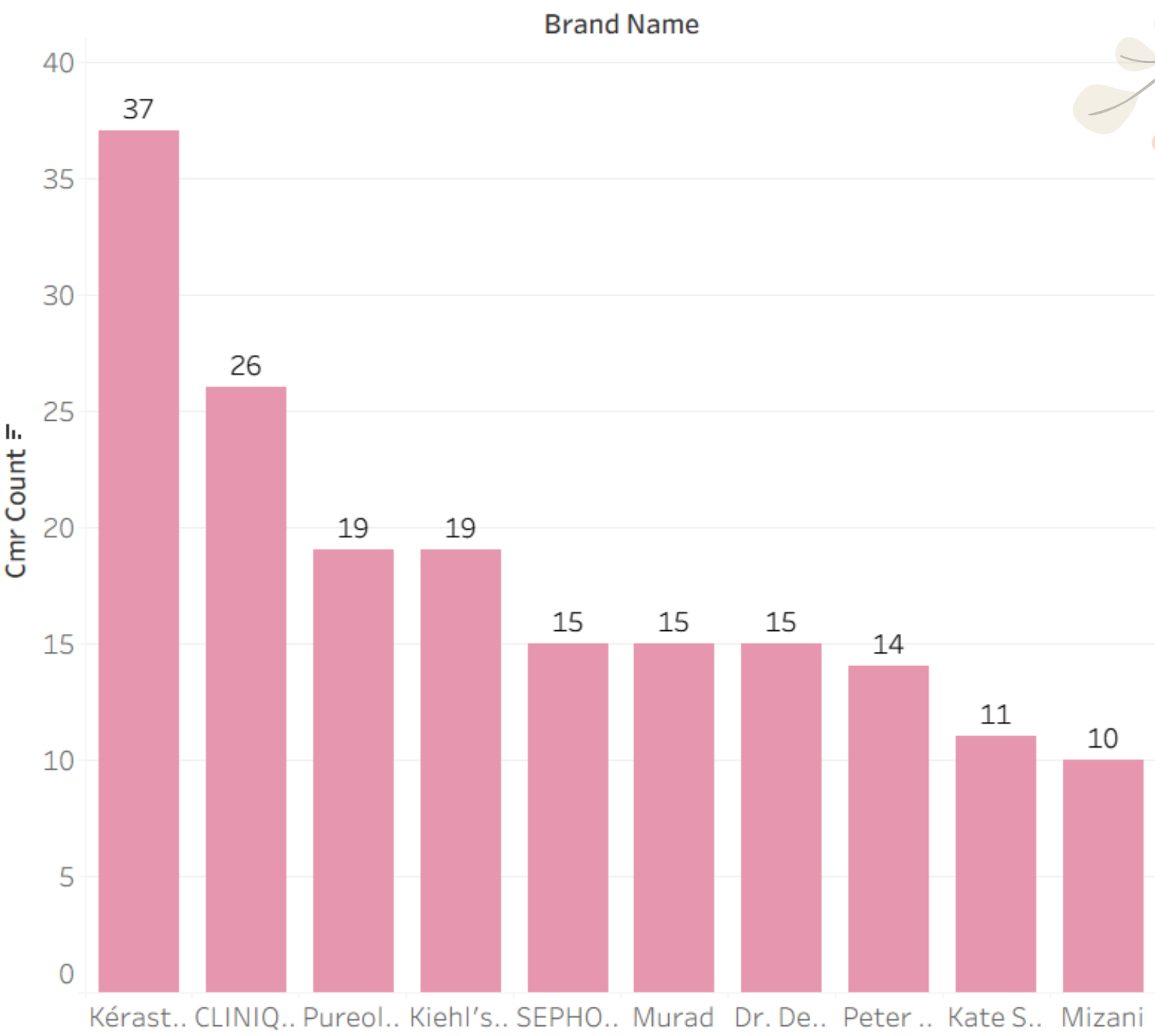


Top 10 Concerned Brands

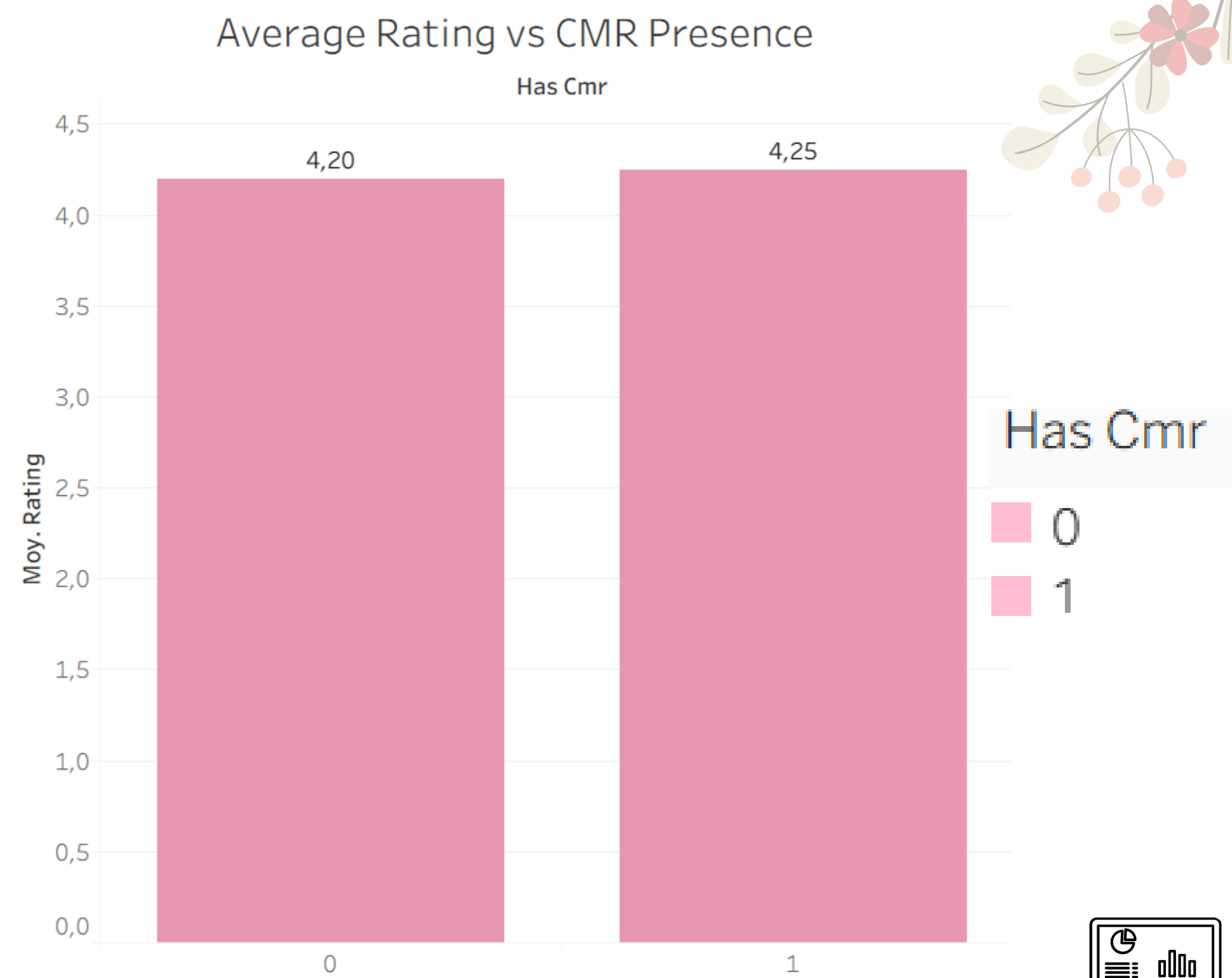
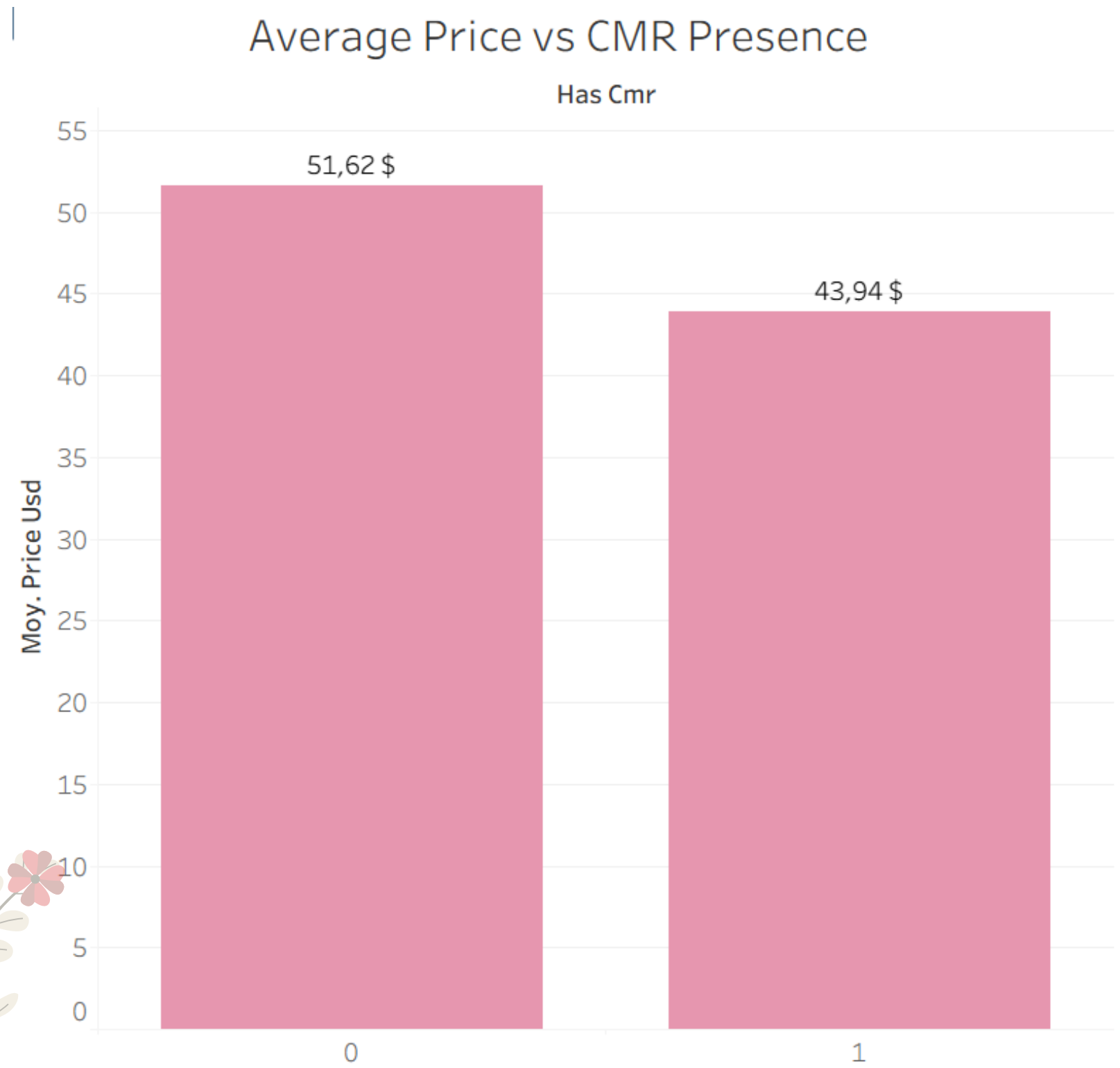
Top Brands – Restricted Ingredients



Top Brands – CMR Risk



Affect On Market Performance?



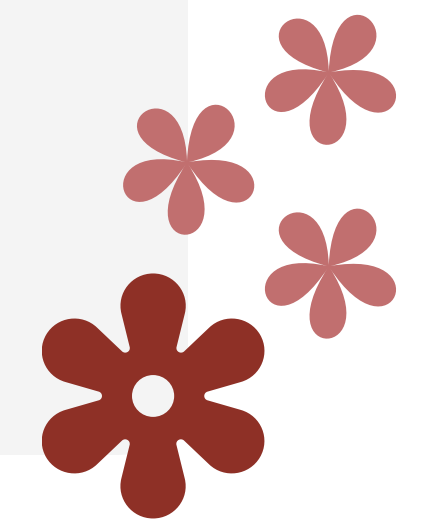
Limitations & Future Improvements



Limitations:

- Ingredient concentration data not available
- Regulatory scope limited to Annex III

Future Improvements:

- Expand to other annexes and include ingredients concentrations
 - Machine learning for automated regulatory alerts
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Conclusion

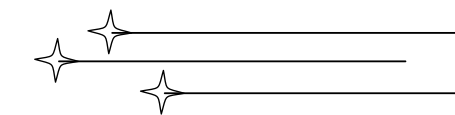
Regulatory risk is not evenly distributed across product categories or brands

CMR substances require immediate attention and represent critical risks

Consumer behavior does not always align with regulatory safety concerns

Market success does not guarantee clean composition





Thank You For Your Attention

Zina BABACI

